

Firmware Footprints - Solution Report

Challenge Summary

Objective: Reverse-engineer the recovered ARM firmware snippet and locate the hardcoded secret used for internal authentication protocols.

Techniques Used

1. ELF File Identification using file/readelf.
2. Static analysis of sections (.text, .rodata, .data).
3. String extraction using strings utility.
4. Symbol enumeration using nm.
5. Memory inspection around authentication-related symbols.

Analysis Findings

The binary is an ELF 32-bit ARM executable (EABI5), not stripped.

Authentication-related artifacts discovered:

```
Internal auth initialized: %s
internal_auth_secret
```

Secret Recovery

The symbol 'internal_auth_secret' pointed to a string stored in the firmware's data/rodata region.

```
Recovered secret:
INT_COMMS_KEY_AE_X789
```

Flag

```
FLAG{INT_COMMS_KEY_AE_X789}
```

Conclusion

Static reverse engineering of the ARM firmware exposed a hardcoded internal communication authentication secret. The recovered value forms the challenge flag.